

Introduction to Probability and Statistics

Solution to Exercises Lecture 5

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Exercise 1: Adapted from Chapter 8, problem 25

It is claimed that a certain type of bipolar transistor has a mean value of current gain that is at least 210. A sample of these transistors is tested. If the sample mean value of the current gain is 200 and it is known that the standard deviation is 35, would the claim be rejected at the 5 percent level of significance if

- a) The sample size is 25?
- b) The sample size is 64?

Tips: First, formulate the null hypothesis H_0 . Then, identify the type of test:

- Is it one-side or two-side test?
- What is the test statistic T that you should use?
- Should you perform a z-test or a t-test?
- Will you compare T to the threshold c or the p-value to α ?

Try to interpret the results.

Solution:

- a) The sample size is 25?

The null hypothesis is $\mu \geq 210$ and we have that the sample mean is $\bar{X}_n = \hat{\mu}_n = 200$ and we know that the standard deviation is $S_n = 35$.

Since we know the standard deviation $\sigma = 35$, we perform a one-sided test using the normal distribution.

Several of the exercises were taken from the book "Introduction to Probability and Statistics for Engineers and Scientists" by Sheldon M. Ross, 3rd Ed.

The test statistic for this type of one-sided test is $T = (\hat{\mu}_n - \mu_0)\sqrt{n}/S_n$ and the rejection region is $R = \{x : T(x) < c\}$. We also need to calculate the value of $c = -Z_\alpha$.

Option 1: We get that $T = -1.4285$ and $c = -1.6449$. Since $T > c$, we **cannot reject** H_0 .

Option 2: We calculate the p-value as $v = \Phi(T) = 0.0766$. Since $v > \alpha$, we **cannot reject** H_0 .

b) The sample size is 64?

Now we have that $n = 64$ and re-do the calculations.

Option 1: We get that $T = -2.2857$ and $c = -1.6449$. Since $T < c$, we **reject** H_0 .

Option 2: We calculate the p-value as $v = \Phi(T) = 0.111$. Since $v < \alpha$, we **reject** H_0 .

Interpretation

In this case, we cannot reject H_0 when the sample size is $n = 25$. This is because, even though the sample mean is lower than the value of $\mu_0 > \hat{\mu}_n$, this outcome is not so extreme with μ_0 and a relatively small sample size of 25 as indicated by the p-value of 0.0766.

As the sample size increases, it would be expected that extreme outcomes in the sample mean happen more rarely because the standard deviation of the sample mean $\sigma/\sqrt{n}$ decreases as n increases. However, this is not the case and the outcome for a sample mean of 200 is considered an extreme case as indicated by the p-value of 0.0111. Therefore, H_0 is rejected for $n = 64$ but not for $n = 25$.

Exercise 2: Chapter 8, problem 56

According to the U.S. Bureau of Census, 25.5 percent of the population of those age 18 or over smoked in 1990. A scientist has recently claimed that this percentage has since increased, and to prove her claim she randomly sampled 500 individuals from this population. If 138 of them were smokers, is her claim proved? Use the 5 percent level of significance.

Tips: Which type of distribution do the RVs have when there are only two options for the outcomes? What is the type of distribution that results from the sum of several of these RVs?

Solution:

The point here is that the scientist wants to find evidence to reject the hypothesis that the ratio of smokers in the U.S. is still 25.5 percent or lesser. So, we make the hypothesis $H_0 : p \leq p_0$,

where $p_0 = 0.255$. We find that the p-value for this hypothesis is

$$v = \Pr(X \geq 138; p_0) = \sum_{k=138}^{500} \binom{500}{k} p_0^k (1 - p_0)^{500-k} = 0.1525. \quad (1)$$

Since $v = 0.1525 > \alpha = 0.05$, the hypothesis H_0 cannot be rejected and we can assume that there has not been an increase in the percentage of smokers.

Another option is using the normal approximation use $\hat{p}_n = 0.276$ and performing a t-test with test statistic $T = (\hat{p}_n - p_0) \sqrt{n} / \sqrt{p_0(1 - p_0)}$. The rejection region is $T > T_{n-1, \alpha}$. The calculated test statistic from the evidence is $T = 1.0773$ and $T_{n-1, \alpha} = 1.6479$, so the null hypothesis cannot be rejected.

Exercise 3: The students from last year were asked to choose between the following two options for the rest of the course using an online poll.

- 1) Israel gives the rest of the theory on March 26. Students do the hypothesis testing exercises on their own. We do the workshop exercises on April 18.
- 2) Israel gives half of the theory on March 26 and the rest on April 18. We do the exercises for hypothesis testing and the workshop mixed with the lectures.

Naturally, not all the students responded to the poll. As Fig. 1 with a total of $n = 32$ responses, 15 of them selecting option 2).

What do you prefer?

32 responses

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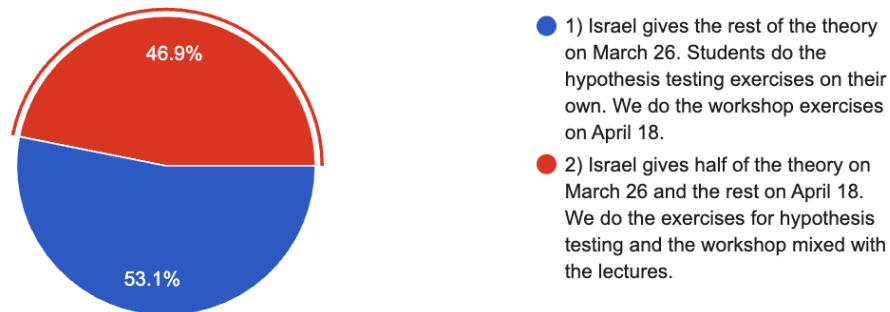


Fig. 1. Results of the voting for the next sessions of statistics.

To make a decision that is fair, Israel decided to conduct a test. First, he defined $X_i \sim \text{Bernoulli}(p)$, where $X_i = 0$ if a student selects 1) and $X_i = 1$ the students selects 2). With this,

he set $p_0 = 0.5$, defined the null hypothesis H_0 : “the students prefer option 2)”, which can be formulated as $p \geq p_0$. Then, he estimated

$$\hat{p}_{32} = \frac{1}{32} \sum_{i=1}^n X_i = 0.469$$

- Perform a one-sided test with $\alpha = 0.05$ using the null hypothesis H_0 and calculating the p-value for the Binomial distribution $v = P(X \leq k; n, p_0)$, where $k = 15$ is the number of students that chose option 2). Can you reject Israel’s hypothesis?
- Change the null hypothesis to be H'_0 : “the students prefer option 1)”. What is the mathematical formulation of this? Can you reject this hypothesis with the collected evidence?
- Assume that the result of the poll is now $k = 11$, meaning that only eleven students preferred option 2). Can the original null hypothesis H_0 be rejected?

Solution:

- We calculate the p-value as

$$v = \Pr(X \leq k; p_0) = \sum_{j=0}^k \binom{n}{j} p_0^j (1 - p_0)^{n-j} = 0.43. \quad (2)$$

Since $v = 0.43 > \alpha = 0.05$, Israel’s hypothesis H_0 cannot be rejected.

- The new null hypothesis to H'_0 : “the students prefer option 1)”, is defined as $p \leq p_0$. Now we calculate the p-value as

$$v = \Pr(X \geq k; p_0) = \sum_{j=k}^n \binom{n}{j} p_0^j (1 - p_0)^{n-j} = 0.7017. \quad (3)$$

Since $v = 0.7017 > \alpha = 0.05$, we cannot reject hypothesis H'_0 either. This means that the evidence is not enough to give a strong statement for rejecting any of the two hypotheses, even though this latter hypothesis has a higher p-value than the previous one.

- We have the same problem for the hypothetical case of $k = 11$. By calculating the p-value $v = \Pr(X \leq k; p_0) = 0.0551$, we observe that is still higher than α . This means that, because the sample size not so large, there are only a few cases where the hypothesis H_0 can be rejected. These cases are those where $k \leq 10$.